

Chapter 13 Programming and Languages

Introduction

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- In this chapter, you focus on Phase 4, Systems Development, of the systems life cycle and learn about the programming process and some of the programming languages that are available
 - Competent end users need to understand the relationship between systems development and programming

Programs and Programming

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- Program
 - A list of instructions for the computer to follow to accomplish the task of processing data into information
 - Statements used in a programming language such as C++, Java, or Python
 - Programs can be
 - Prewritten/packaged
 - Custom-made
 - Programming or Software Development
 - Actually a problem-solving procedure
 - List of instructions for the computer to follow to process data
 - Follows a six-step process known as the System Development Life Cycle

Six-Step Software Development Life Cycle

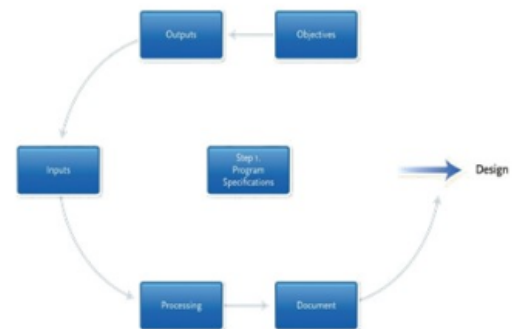
The six steps are as follows:

1. Program specification
2. Program design
3. Program code
4. Program test
5. Program documentation
6. Program test



Program Specification

- Also called program definition or program analysis
- Five items must be specified:
 1. Program's objectives
 2. Desired output
 3. Input data required
 4. Processing requirements
 5. Documentation



Defining Program Specification

- Program objectives
 - Requires a clear statement of the problem being addressed
- Desired output
 - The end-user should communicate the inputs and outputs
- Input data
 - Determine the source of the data
- Processing requirements
 - Tasks to move input to output
- Program specification document
 - Document program specifications

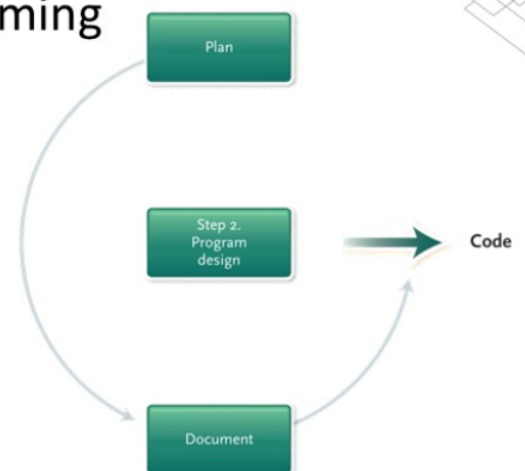
Client name: Allen Realty			Month and Year: Jan. '14	
Date	Worker	Regular Hours & Rate	Overtime Hours & Rate	Bill
1/2	M. Jones	5 @ \$10	1 @ \$15	\$65.00
	K. Williams	4 @ \$30	2 @ \$45	\$210.00

Daily Log			
Worker:			
Date:			
Client	Job	Time in	Time out
A	TV commercial	800	915
B	Billboard ad	835	1200
C	Brochure	1215	1545
D	Magazine ad	1600	1745

Program Design

Plan a solution using structured programming techniques

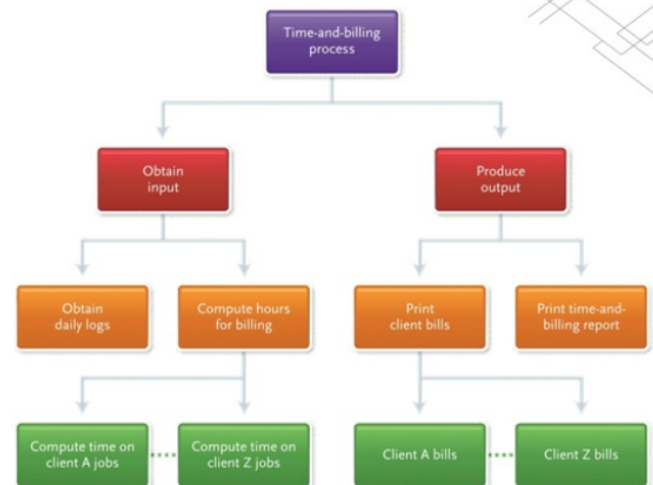
- Techniques
 - Top-down design
 - Pseudocode
 - Flowcharts
 - Logic structures



Top-Down Program Design

To identify the program's processing steps; called modules

- Each module is made up of logically related program statements



Pseudocode

An outline of the logic of the program you will write

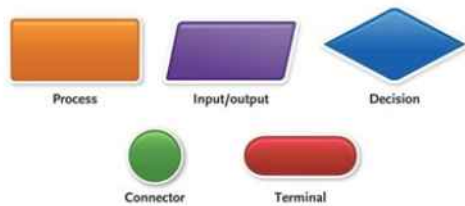
- Summary of the program before it is written

Compute time for Client A

Set total regular hours and total overtime hours to zero.
Get time in and time out for a job.
If worked past 1700 hours, then compute overtime hours.
Compute regular hours.
Add regular hours to total regular hours.
Add overtime hours to total overtime hours.
If there are more jobs for that client, go back and compute for that job as well.

Flowcharts

Graphically present the detailed sequence of steps needed to solve a programming problem



Logic Structures

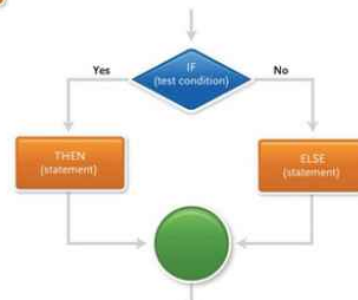
Enables you to write structured programs, which take much of the guesswork out of programming

- Sequential structure
 - One program statement follows another
- Selection structure
 - A decision must be made
- Repetition structure or Loop Structure
 - Describes a process that may be repeated as long as certain condition remains true

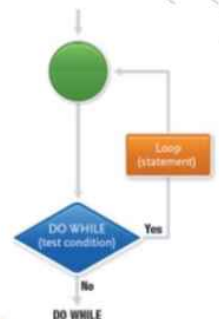
Sequence



Selection



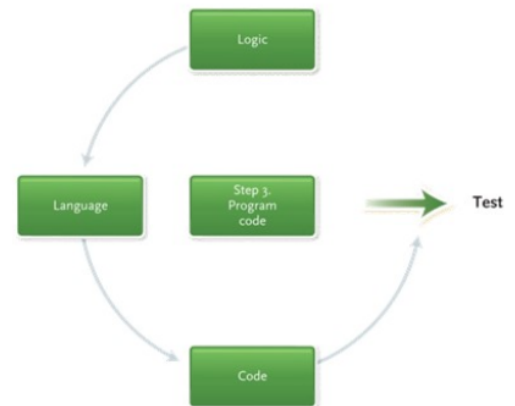
Repetition



Program Code

Writing the program is called coding

- Characteristics of a good program
 - Reliable
 - Produces the correct output
 - Catches common input errors
 - Well-documented and understandable
 - Structured programs – one of the best ways to code effective programs
 - Using logic structure



Coding

Write the program

- A programming language uses a collection of symbols, words, and phrases that instruct a computer to perform specific operations

```
#include <fstream.h>
void main (void)
{
    ifstream input_file;

    float total_regular, total_overtime, regular, overtime;
    int hour_in, minute_in, hour_out, minute_out;
    input_file.open("time.txt", ios::in);

    total_regular = 0;
    total_overtime = 0;

    while (input_file != NULL)
    {
        input_file >> hour_in >> minute_in >> hour_out >> minute_out;
        if (hour_out > 17)
            overtime = (hour_out - 17) * (minute_out / (float)60);
        else
            overtime = 0;
        regular = ((hour_out - hour_in) * (minute_out - minute_in) / (float)60) - overtime;
        total_regular += regular;
        total_overtime += overtime;
    }

    cout << "Regular: " << total_regular << endl;
    cout << "Overtime: " << total_overtime << endl;
}
```


Programming Languages

Language	Description
C++	Extends C to use objects or program modules that can be reused and interchanged between programs
C#	A programming language designed by Microsoft to extend C++ for developing applications in the Windows environment.
Java	Primarily used for Internet applications; similar to C++; runs with a variety of operating systems
JavaScript	Embedded into web pages to provide dynamic and interactive content
Python	General-purpose programming language that is simple and easy to learn. Frequently used in introductory programming courses.
Swift	Uses graphical user interface and special code for touch screen interfaces to create apps for Apple iOS devices

Program Test

- Debugging
 - The process of testing and then eliminating errors such as:
 - Syntax errors are a violation of the rules of programming language
 - Logic errors occur when the programmer uses incorrect calculation or leaves out a programming procedure
 - Testing process involves one or more of several methods



The screenshot shows a code editor window with a C++ program. The code is as follows:

```
#include <iostream>
using namespace std;

int main()
{
    int total_regular, total_overtime, regular_overtime;
    int hour_in, minute_in, hour_out, minute_out;
    int file_open("time.txt", ios::in);
    total_regular = 0;
    total_overtime = 0;
    while (file_open != NULL)
    {
        input_file << hour_in << minute_in << hour_out << minute_out;
        if (hour_out <= 17)
            overtime = (hour_out - 17) * (minute_out / 1500);
        else
            overtime = (hour_out - 17) * (minute_out / 1500) + (minute_out % 1500) / 1500;
    }
    cout << "Compute time (input) = " << total_regular << " minutes\n";
    cout << "Compute time (overtime) = " << overtime << " minutes\n";
    return 0;
}
```

A red arrow points to the line `cout << "Compute time (input) = " << total_regular << " minutes\n";` with the message "Syntax error". The error message at the bottom of the window reads: "Error: unexpected 'n' in string literal, expected ';'".

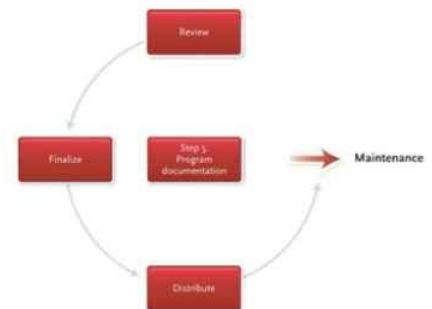
Program Testing Processes

- Desk checking or Code review
 - Printout of program reviewed line by line
- Manual testing with sample data
 - Correct and incorrect data manually entered, results evaluated
- Attempt at translation
 - Written program goes through translator program on the computer, must be syntax error free
- Testing sample data on the computer
 - Tests for logic errors
- Beta testing
 - Testing by a select group of potential users; users provide feedback

Program Documentation

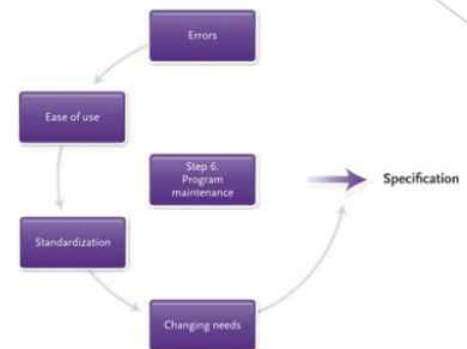
Written descriptions and procedures about a program and how to use it

- Carried on throughout the programming steps
- Important for people who will use and/or support the program
 - Users need to know how to use the software
 - Operators need to know what to do about any error messages
 - Programmers may even remember all the details
 - Those taking over the program will need to know details



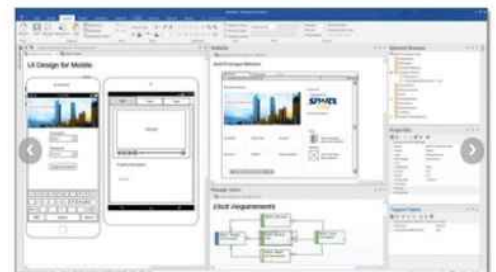
Program Maintenance

- 75% of total lifetime cost for an application is for maintenance
- Ensures program is
 - Error-free
 - Effective
 - Efficient
- Two activity categories
 1. Operations
 - Patches – programming modification or corrections
 - Software updates – significant patches
 2. Changing needs
 - Agile development – starts with getting the core functionality working then expands through to customer satisfaction



Case Tools and Object-Oriented Programming

- Computer-aided software engineering (CASE)
 - Automates portions of the development process
- Object-oriented software – OOP
 - Focuses less on procedures, more on relationships with previously defined procedure
 - Objects contain both the data and the processing operations needed to perform a task



Generations of Programming Languages

- Levels or Generations
 - Coding from machine languages to human or natural languages
- There are five distinct generations
 - Lower level is closer to machine language
 - Higher level is closer to human-like language

5 Generations

- 1st Gen: Machine languages
 - Data represented in 1s and 0s
- 2nd Gen: Assembly languages
 - Uses abbreviations or mnemonics that are automatically converted to the appropriate sequence of 1s and 0s
- 3rd Gen: High level procedural languages (3GLs)
 - Designed to express the logic – the procedures – that can solve general problems. Translated into machine language with a compiler or an interpreter
- 4th Gen: Task-oriented languages (4GLs)
 - Designed to solve specific problems
- 5th Gen: Problem and Constraint languages (5GL)
 - Computer languages that incorporate the concepts of artificial intelligence to allow a person to provide a system with a problem and some constraints and then request a solution

Generation Samples

- Listed to the right are samples of each generation programming language in order from the 1st through the 5th

```
111100100111001111010010000100000111000000101011
```

```
ADD 210(8,13),02B(4,7)
```

```
if (score >= 90) grade = 'A';
```

```
SELECT client FROM dailyLog WHERE serviceEnd > 17
```

```
Get patientDiagnosis from patientSymptoms "sneezing", "coughing",  
"aching"
```

Careers In IT

- Computer programmers create, test, troubleshoot, update and repair programs
- Employers seek individuals with
 - Degree in Computer Science or Information Systems
 - Desired traits include patience, logical thinking, and attention to detail
- Computer Programmers can expect to earn \$49,000 to \$78,000 annually



A Look to the Future ~ Your Own Programmable Robot

- It may not be to much longer before you have your own robot
- Hardware components needed to create are becoming cheaper

